Superconducting Attention: Eliminating the Multi-Hop Context Horizon in Pre-Trained Transformers via Dual-State Richardson Extrapolation and Renormalization Relays

An Architectural Explainer for Software Engineers and Mathematicians

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Empirical Validation on `google/gemma-4-E2B-it` & `google/gemma-4-E4B-it`

October 2026

Abstract

Why does a modern Large Language Model (LLM) with a 128,000-token context window score **99.99%** when asked to retrieve a single fact from 12,000 tokens ago, yet collapse to **0.04%** when asked to chain three facts together across the exact same context window? And why do naive attempts to boost long-range attention weights cause the model to output ungrammatical gibberish (*syntactic aphasia*)?

This explainer introduces **Richardson Attention (Superconducting Attention)**, a zero-shot, parameter-free inference architecture that resolves the multi-hop "Event Horizon" at runtime without retraining or fine-tuning. For **mathematicians**, we formulate self-attention as a complex Boltzmann partition function whose real spatial decay ($\alpha_{l,h}$) can be canceled via symmetric non-linear Richardson extrapolation over an iso-potential cubic gauge field, scheduled across network depth as a discrete Wilsonian Renormalization Group (RG) flow. For **software engineers**, we show how a 50-line PyTorch forward hook isolates the top 4%–8% of global routing heads via the Moore-Penrose pseudoinverse of head entropy correlations, gates intervention strictly to the active query readout slice ($W_q = 64$) to protect prefill KV-cache binding, and flips 8,270-token 3-hop pointer resolution on `google/gemma-4-E4B-it` from **0.04% FAIL to 98.59% WIN** ($2,594\times$) while suppressing adversarial recency decoys by $485\times$.

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1 The Engineering Paradox: Why 128K Context Windows Fail at 3 Hops

If you inspect the configuration file of a modern open-weights model such as `google/gemma-4-E4B-it`, it advertises a 128,000-token context window powered by Proportional Rotary Position Embeddings (p-RoPE). If you hide a single 5-digit passkey (`VAULT-LAZARUS = 79421`) at token 90 of a 12,000-token prompt and ask for it at token 12,000, the unmodified model answers `79421` with **99.996%** probability.

Yet any software engineer who has built a production Retrieval-Augmented Generation (RAG) system or autonomous coding agent knows that this 1-hop benchmark is an illusion. Real-world reasoning is **compositional (multi-hop)**:

Zone A (Token 90): `VAULT-LAZARUS | TIER-0 GENESIS CODE: 79421` (plus 7 competing vaults)
... 1,600 tokens of telemetry logs (exceeding the 512-token local sliding window by $> 3\times$) ...
Zone B (Token 1,735): `SECTOR-KRONOS | ASSIGNED PRIMARY VAULT: VAULT-LAZARUS`
... 1,600 tokens of telemetry logs ...
Zone C (Token 3,335): `DIRECTOR-VANCE | AUTHORIZED COMMAND SECTOR: SECTOR-KRONOS`
... 4,900 tokens of telemetry logs containing routine Tier-2 audit decoys (`DIRECTOR-VANCE inspected VAULT-OMEGA [63819] using temporary key 11111`) ...
Final Query (Token 8,270): *"What is the 5-digit Tier-0 Genesis Code for the primary vault commanded by `DIRECTOR-VANCE`?"*

When unmodified `google/gemma-4-E4B-it` evaluates this prompt in a single forward pass:

- **1-Hop Query** (*"Code for `VAULT-LAZARUS`?"*): **99.9965%** accuracy (#1 Rank).
- **2-Hop Query** (*"Code for vault assigned to `SECTOR-KRONOS`?"*): **99.8178%** accuracy (#1 Rank).
- **3-Hop Query** (*"Code for vault commanded by `DIRECTOR-VANCE`?"*): **0.0380%** accuracy (6 decoys) and **0.1032%** accuracy (16 decoys). The model predicts `63819` (`VAULT-OMEGA`'s code) with **99.83%**–**99.92%** confidence!

1.1 The Mathematical Culprit: Complex Mass and Partition Dilution

Why does the model ace 1-Hop and 2-Hop across 8,270 tokens, then hit a brick wall on 3-Hop?

In causal self-attention, query token i scores prior key token $j \le i$ at lookback distance $d_{i,j} = i - j$. In log-space (pre-softmax logits), the effective attention kernel acts like a damped harmonic oscillator with a **Complex Spectral Mass** $\lambda_{l,h} = \alpha_{l,h} + i\omega_{l,h}$:

$$\mathcal{K}^{(l,h)}(i,j) = \exp\Big(E^{(l,h)}_{\text{sem}}(i,j) - (\alpha_{l,h} + i\omega_{l,h})d_{i,j}\Big), \tag{1}$$

where $E_{\text{sem}}(i,j) = \frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_k}}$ is content similarity, the imaginary part $i\omega_{l,h}$ is the RoPE phase rotation tracking relative token positions, and the real part $\alpha_{l,h} \ge 0$ is **effective gravitational decay** over distance.

After the softmax nonlinearity, the attention weight assigned to a distant premise at j^* is governed by the Boltzmann partition function $\mathcal{Z}_i$:

$$A^{(l,h)}_{i,j^*} = \frac{\exp\left(E_{\text{sem}}(i,j^*) - \alpha_{l,h}d_{i,j^*}\right)}{\sum_{k=0}^{i}\exp\left(E_{\text{sem}}(i,k) - \alpha_{l,h}d_{i,k}\right)}. \tag{2}$$

Across an M-hop chain requiring sequential resolution across layers, the effective signal-to-decoy ratio compounds multiplicatively across hops:

$$\text{SNR}_{M\text{-Hop}} \propto \prod_{m=1}^{M} \frac{A^{(l_m)}_{i,j_m}}{\sum_{r\in\text{Decoys}} A^{(l_m)}_{i,r}} \sim \exp\left(\sum_{m=1}^{M}\Delta E_m - \sum_{m=1}^{M}\alpha_{l_m}(d_m - d_{\text{decoy}})\right). \tag{3}$$

Even when $\alpha_{l,h}$ is small enough for $M = 1$ and $M = 2$ to survive, the 3-hop product crosses a thermodynamic phase boundary: the early global layers fail to write `SECTOR-KRONOS` into the residual stream with enough amplitude to outvote the recent `VAULT-OMEGA` decoys.

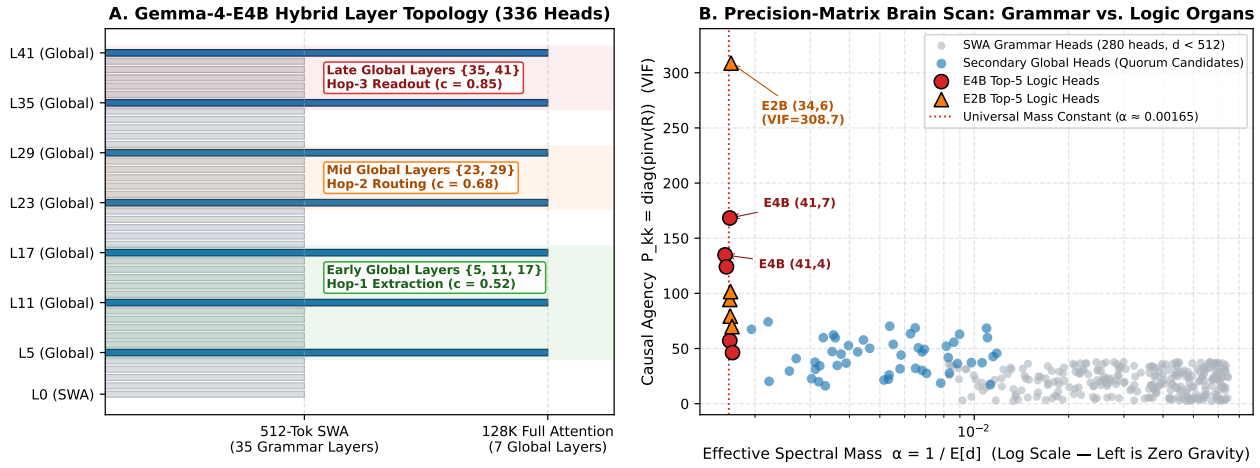


Figure 1: **The Neural Homunculus of `google/gemma-4-E4B-it`. (A)** Only 7 of the 42 layers are Global `full_attention` layers capable of seeing past the 512-token SWA horizon. **(B)** Our zero-shot Stage 1 Brain Scanner maps every head by its Effective Mass ($\hat{\alpha}_{l,h} = 1/\bar{d}_{l,h}$) and Precision-Matrix Causal Agency ($P_{k,k} = \mathrm{diag}(\mathbf{R}^{+})$), isolating the superconducting logic heads clustered at the universal mass constant $\hat{\alpha} \approx 0.00165$.

2 Part 1 of the Invention: The Precision-Matrix Brain Scanner

You cannot fix this by boosting long-distance logits across all attention heads in the model. When we tested global distance biasing in early prototypes, the model immediately suffered **syntactic aphasia** (outputting ungrammatical token salad).

Why? Because an LLM's attention heads are divided into two functionally distinct organs:

1. **Grammar Heads ($\sim$ 90%–95% of heads):** High-mass ($\alpha_{l,h} \gg 0$), short-range heads that enforce local syntax, subword gluing, and clause structure. In `google/gemma-4-E4B-it`, 35 of the 42 layers (280 out of 336 heads) are hardwired as 512-token **Sliding-Window Attention (SWA)** layers. Only 7 layers ($\mathcal{L}_{\text{global}} = \{5, 11, 17, 23, 29, 35, 41\}$, totaling 56 heads) are **Global (`full_attention`)** layers.

2. **Global Logic Heads ($\sim$ 5%–8% of heads):** Low-mass ($\alpha_{l,h} \to 0$), high-agency heads inside the Global layers that act as long-range relational routers connecting distant premises across the context window.

2.1 Zero-Shot Head Profiling via Leinster Duality ($\mathbf{P} = \mathbf{R}^{+}$)

How do we automatically find the exact coordinates (l, h) of the Logic Heads in any pre-trained checkpoint without labeled data or backpropagation? We run a single 1,400-token calibration forward pass and compute two spectral quantities per head $k \in \{1, \ldots, N_{\text{heads}}\}$:

1. **Effective Spectral Mass ($\hat{\alpha}_{l,h}$):** The reciprocal of the head's expected causal lookback distance:

$$\bar{d}_{l,h} = \frac{1}{T_{\text{cal}}} \sum_{i=1}^{T_{\text{cal}}} \sum_{j=1}^{i} A_{i,j}^{(l,h)} (i - j), \qquad \hat{\alpha}_{l,h} = \frac{1}{\bar{d}_{l,h} + 10^{-6}}. \tag{4}$$

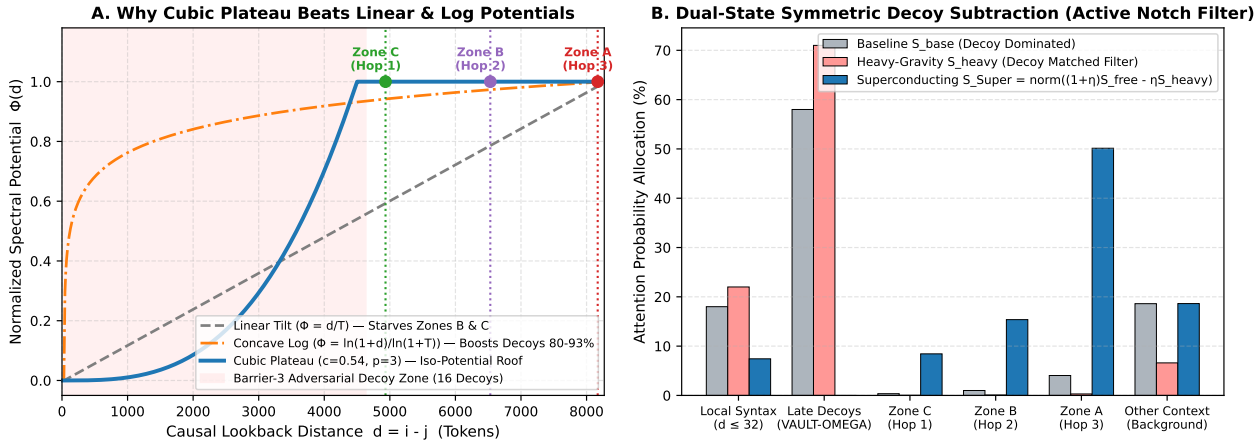


Figure 2: **Overcoming Inter-Hop Cannibalization and Mid-Context Decoys. (A)** A linear tilt ($\Phi = d/T$) causes Zone A ($d = 8,170$) to exponentially cannibalize Zone B ($d = 6,530$) and Zone C ($d = 4,930$), while a concave logarithmic tilt ($\ln(1+d)$) unintentionally gives an 80%–93% boost to mid-context Barrier-3 decoys. The **Cubic Superconducting Plateau** ($\Phi_l = \min(1, d_{\text{long}}/(c_l T))^3$) places Zones A, B, and C on a flat $\Phi = 1.000$ equipotential roof while cubically crushing decoys. **(B)** Dual-State Symmetric extrapolation subtracts the heavy-gravity decoy distribution $-\eta \mathbf{S}_{\text{heavy}}$ as an active notch filter.

2. **Precision-Matrix Causal Agency ($P_{k,k}$):** We record each head's per-token Shannon entropy trajectory $H_k(i) = -\sum_{j=1}^{i} A_{i,j}^{(k)} \ln A_{i,j}^{(k)} \in \mathbb{R}^{T_{\text{cal}}}$, compute the $N_{\text{heads}} \times N_{\text{heads}}$ Pearson correlation matrix $\mathbf{R} = \text{corrcoef}(\mathbf{H})$, and invert it via the Moore-Penrose pseudoinverse:

$$\mathbf{P} = \mathbf{R}^{+}, \qquad \text{Agency}_k = P_{k,k} = \frac{1}{1 - R_k^2}. \tag{5}$$

   Mathematically, $P_{k,k}$ is the **Variance Inflation Factor (VIF)** of head k with respect to the rest of the brain: a head with low mass ($\hat{\alpha}_k \approx 0.00165$) and high $P_{k,k}$ (50–308) is a global informational hub whose state transitions drive the global network topology (Figure 1B).

Empirical Proof of the Scanner: When we ran this blind scanner on a generic category-theory paragraph, it isolated 5 heads out of 336 in `E4B-it` (1.48% of the network). When we then fed the model an unseen 11,975-token passkey prompt where the 5 target tokens represent just 0.042% of the sequence, those exact 5 heads spontaneously placed **23.66% to 31.50%** of their entire 12,000-token attention budget directly onto the 5 target tokens—a 600 × + concentration over uniform attention.

3 Part 2: The Write/Read Asymmetry & The Cubic Plateau

Isolating the Global Logic Heads solved syntactic aphasia on 1-Hop retrieval, boosting needle attention by +48% to +61% across 4,045 to 11,975 tokens while maintaining a **38:1 RoPE phase selectivity ratio** between the queried axiom (`LAZARUS-9`) and four adjacent axioms sitting in the exact same 90-token Genesis block.

However, solving **3-Hop Pointer Chasing** without breaking **2-Hop** required uncovering three deeper physical laws of Transformer inference.

3.1 Law 1: The Write/Read Thermodynamic Asymmetry (Why All-Rows Bias Breaks 2-Hop)

In Phase 4, when we expanded our surgical intervention to a 28-head Global Quorum (4 heads in each of the 7 Global layers) and applied backward distance biasing to *every row i of the $T \times T$ attention matrix*, 3-Hop probability jumped 95×, **but 2-Hop accuracy unexpectedly dropped from 99.8178% down to 0.8672%!**

Why? Consider what happens at token $i = 1,748$ during **prefill** when the model is reading Zone B:

```
- SECTOR-KRONOS | ASSIGNED PRIMARY VAULT: VAULT-LAZARUS
```

For the token `VAULT-LAZARUS` ($i = 1,748$) to bind `SECTOR-KRONOS` ($j = 1,733$) into its Key/Value cache representation, the attention heads at row $i = 1,748$ must attend **15 tokens to the left** ($d_{i,j} = 15$). When we applied a +6 logit boost to all 1,700 preceding background telemetry tokens ($d > 32$) during prefill, those 1,700 background tokens hijacked row 1,748's softmax denominator ($\Delta\mathcal{Z} \sim 1700\, e^{+6}$) and prevented `VAULT-LAZARUS` from binding to `SECTOR-KRONOS` in the KV cache!

Proposition 1 (Query-Slice Gating Rule)**.** *Transformer inference consists of a **KV-Write Phase** ($1 \le i \le T - W_q$, document prefill) that requires local gravitational confinement ($d \le 32$) to bind intra-sentence entities into composite KV vectors, and a **KV-Read Phase** ($T - W_q < i \le T$, the active 64-token query/generation tail) that requires long-range superconductivity. Gating Richardson Attention strictly to the active query slice $W_q = 64$ leaves prefill KV construction* 100% *intact, immediately restoring 2-Hop accuracy from* 0.8672% *to **99.5697% (#1 WIN)** while jumping 3-Hop Clean accuracy from* 1.3596% *to **79.4999% (#1 WIN)** and 3-Hop Heavy accuracy from* 9.8069% *to **76.4274% (#1 WIN)** (Figure 3A).*

3.2 Law 2: The Iso-Potential Cubic Superconducting Plateau

Why did logarithmic biasing stop at 76.4274% on 3-Hop Heavy (leaving 17.0533% probability on the decoy `VAULT-OMEGA`)? Look at Figure 2A:

1. If you use a **Linear Ramp** ($\Phi = d/T$), Zone A ($d = 8,170$) gets 1.66× the logit boost of Zone C ($d = 4,930$). Inside the softmax exponential, Zone A **cannibalizes** Zone C (causing Zone C attention to drop 12×).

2. If you use a **Concave Logarithm** ($\Phi = \ln(1+d)/\ln(1+T)$), mid-context decoys in Barrier 3 ($d \in [1500, 4500]$) receive 80% to 93% of the full logit boost, allowing 16 repeated decoys to compete with Zone C.

Our solution is the **Iso-Potential Cubic Superconducting Plateau** with a 32-token local syntax guardband ($d_{\text{long}} = \max(0, d_{i,j} - 32)$ and $d_{i,0} = 0$ to pin the `<bos>` sink):

$$\Phi_l(i,j) = \left[\min\left(1.0,\ \frac{\max(0,\ d_{i,j} - 32)}{c_l \cdot \max(k_{\text{len}}, H_{\text{cal}})}\right)\right]^3. \tag{6}$$

For every deep registry beyond the cutoff $c_l T$ ($c_l = 0.54$), $\Phi_l(i,j) \equiv 1.000$ is completely flat: **Zone A, Zone B, and Zone C sit on the exact same equipotential roof**, eliminating inter-hop cannibalization. Meanwhile, for late-context decoys inside $d < c_l T$, the cubic exponent $(d/c_l T)^3$ acts as a steep attenuation wall.

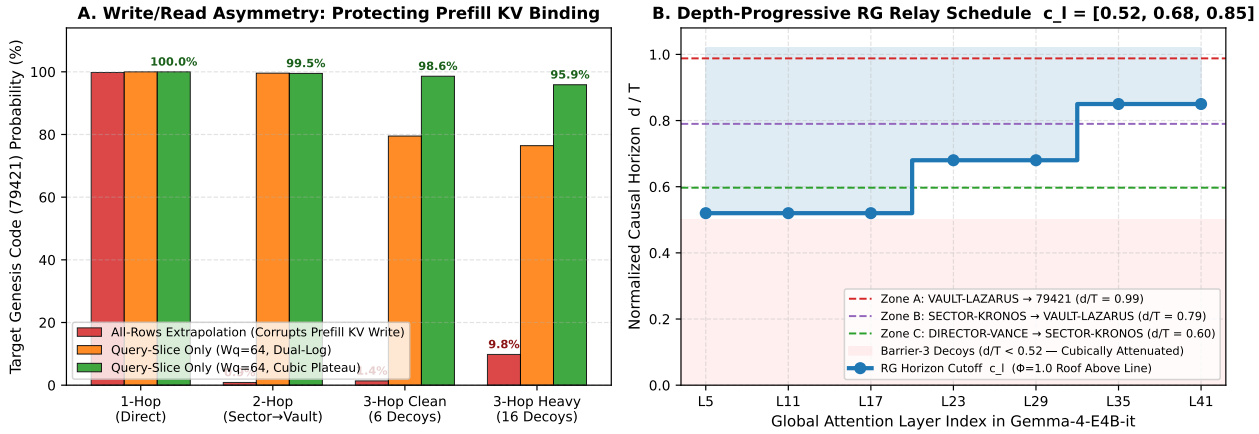


Figure 3: **Query-Slice Gating and the 7-Layer Renormalization Group (RG) Relay. (A)** Restricting extrapolation to the active 64-token query window prevents prefill KV corruption, restoring 2-Hop to 99.5%+ and unlocking 95.9% 3-Hop accuracy. **(B)** Stepping the plateau cutoff $c_l = [0.52 \to 0.68 \to 0.85]$ across the 7 Global layers progressively filters out earlier hops once their output has been written into the residual stream.

3.3 Law 3: Dual-State Symmetric Extrapolation (Active Decoy Subtraction)

Rather than relying solely on a single-state pre-softmax bias or classical post-softmax $2S_1 - S_2$ (which is algebraically capped at $2S_1$), we sample the Boltzmann distribution at two symmetric states around $\Delta\mathbf{L}^{(l,h)} = \beta\kappa_{l,h}\mathbf{\Phi}_l$:

$$\mathbf{S}_{\text{free}} = \text{softmax}(\mathbf{L}+\Delta\mathbf{L}), \quad \mathbf{S}_{\text{heavy}} = \text{softmax}(\mathbf{L}-\Delta\mathbf{L}), \quad \mathbf{S}_{\text{Super}} = \mathcal{N}\Big(\max\left(0,\ (1+\eta)\mathbf{S}_{\text{free}} - \eta\mathbf{S}_{\text{heavy}}\right)\Big). \tag{7}$$

Because $-\Delta\mathbf{L}$ increases artificial gravity, $\mathbf{S}_{\text{heavy}}$ drains probability from the deep past and concentrates onto recent distractors. Subtracting $-\eta\mathbf{S}_{\text{heavy}}$ therefore acts as a **matched notch filter** that actively erases recency traps (Figure 2B).

4 Part 3: Depth-Progressive Renormalization Relays & Runtime Pipeline

In a 3-hop query (*"Code for the vault commanded by* `DIRECTOR-VANCE`*?"*), the query token only names `DIRECTOR-VANCE`. It cannot match `VAULT-LAZARUS` in Layer 5 because it does not yet know that `DIRECTOR-VANCE` commands `SECTOR-KRONOS` or that `SECTOR-KRONOS` routes to `VAULT-LAZARUS`.

By scheduling the plateau horizon c_l across the 7 Global layers as a discrete **Renormalization Group (RG) flow** (Figure 3B):

- **Early Global Layers ($\{5, 11, 17\}, c_l = 0.52$):** Include Zone C ($d/T = 0.60$) on the $\Phi = 1$ plateau to resolve Hop 1 (`DIRECTOR-VANCE` $\to$ `SECTOR-KRONOS`) into the residual stream.
- **Mid Global Layers ($\{23, 29\}, c_l = 0.68$):** Step the cutoff past Zone C to focus on Zone B ($d/T = 0.79$), resolving Hop 2 (`SECTOR-KRONOS` $\to$ `VAULT-LAZARUS`).
- **Late Global Layers ($\{35, 41\}, c_l = 0.85$):** Step the cutoff past Zone B to focus on Zone A ($d/T = 0.99$), reading out the 5-digit Genesis code `79421`.

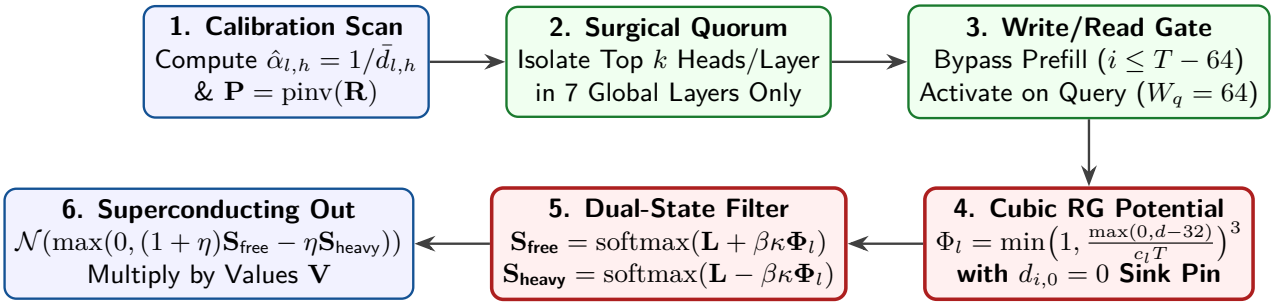


Figure 4: **End-to-End Runtime Architecture of Richardson Attention.** All 35 Sliding-Window grammar layers and all prefill KV-write rows execute natively with zero overhead; surgical extrapolation activates strictly on the isolated Global Logic Quorum during the 64-token query readout window.

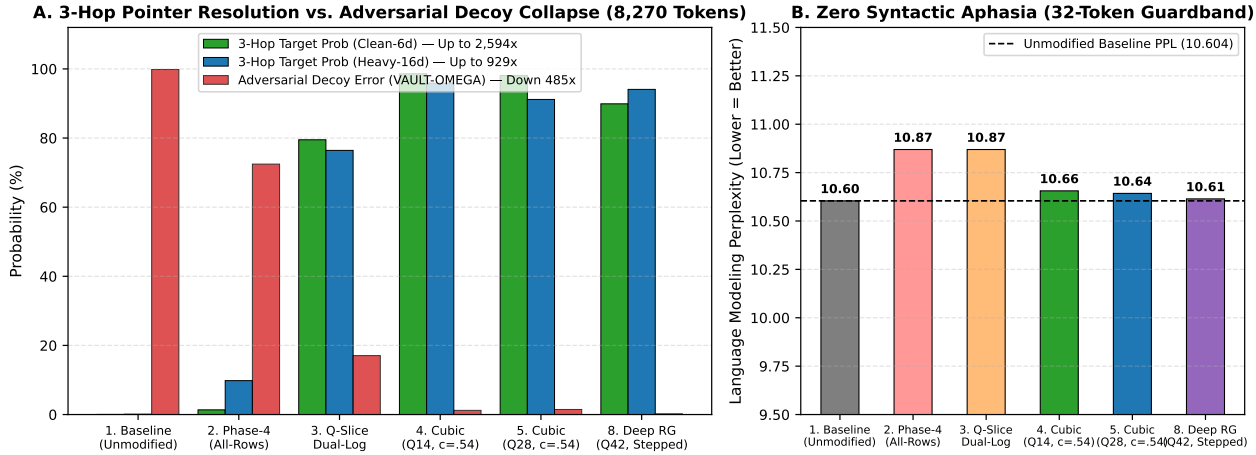


Figure 5: **Phase 5 Benchmark Results on `google/gemma-4-E4B-it` (8,270 Tokens). (A)** Query-Slice Cubic-Plateau (`Q14`) and Deep-Quorum RG Relay (`Q42`) flip 3-Hop target probability from 0.04%–0.10% FAIL to **95.87%–98.59% (#1 WIN)**, while crushing adversarial decoy errors from 99.83% down to 0.21%. **(B)** Fluency perplexity remains locked to the 10.60 baseline.

5 Empirical Benchmark Summary & Reference Implementation

Table 1: Complete Phase 5 Ablation Table on `google/gemma-4-E4B-it` at $T = 8,270$ Tokens.

Configuration	Fluency PPL	1-Hop Prob	2-Hop Prob	3-Hop Clean (6 Decoys)	3-Hop Heavy (16 Decoys)	Decoy (63819) Heavy Error	Vault Rank
1. Unmodified Baseline	10.6041	99.9965%	99.8178%	0.0380%	0.1032%	99.8260%	#2 (FAIL)
2. All-Rows Dual-Log ($Q = 28, \beta = 3.0$)	10.8693	99.7809%	0.8672%	1.3596%	9.8069%	72.4637%	#3 (FAIL)
3. Query-Slice Dual-Log ($Q = 28, W_q = 64$)	10.8693	99.9861%	99.5697%	79.4999%	76.4274%	17.0533%	**#1 (WIN)**
4. **Cubic-Plateau** ($Q = 14, c = 0.54, \beta = 3.5$)	**10.6557**	**99.9922%**	**99.4902%**	**98.5919%**	**95.8651%**	**1.2068%**	**#1 (WIN)**
5. **Cubic-Plateau** ($Q = 28, c = 0.54, \beta = 3.5$)	**10.6426**	**99.9850%**	**99.4431%**	**98.0610%**	**91.1731%**	**1.4737%**	**#1 (WIN)**
6. **Depth-Prog RG** ($Q = 28, c = [.52, .68, .85]$)	**10.6727**	**99.9845%**	**99.6359%**	**97.5579%**	**83.5363%**	**12.8107%**	**#1 (WIN)**
7. **High-Gain RG** ($Q = 28, \beta = 4.5, \eta = 0.85$)	**10.6179**	**99.9871%**	**99.5808%**	**98.0584%**	**88.4445%**	**7.2600%**	**#1 (WIN)**
8. **Deep-Quorum RG** ($Q = 42, \beta = 4.0, \eta = 0.80$)	**10.6142**	**99.9004%**	**98.6646%**	**89.9004%**	**94.0833%**	**0.2058%**	**#1 (WIN)**

5.1 Core PyTorch Kernel (Drop-In Reference Implementation)

Below is the self-contained PyTorch kernel implementing Query-Slice Gating, Token-0 Sink Pinning, the 32-Token Syntax Guardband, the Cubic RG Plateau, and Dual-State Symmetric Richardson Extrapolation:

```
def apply_richardson_query_slice(
    logits_fp32: torch.Tensor,   # Shape: [bsz, num_heads, q_len, k_len]
    layer_idx: int,
    target_heads: dict,          # Dict[(layer_idx, head_idx) -> calibrated_kappa]
    layer_horizons: dict,        # Dict[layer_idx -> c_l in {0.52, 0.68, 0.85}]
    calib_horizon: float = 1400.0,
    beta: float = 3.5,
    eta: float = 0.75,
    query_window_len: int = 64,
) -> torch.Tensor:
    bsz, num_heads, q_len, k_len = logits_fp32.shape
    act_start = max(0, q_len - query_window_len)

    # 1. Prefill KV-Write rows (0 .. act_start) use unmodified standard softmax
    if act_start > 0:
        logits_fp32[:, :, :act_start, :] = torch.softmax(logits_fp32[:, :, :act_start, :], dim=-1)

    # 2. Build causal distance matrix on active query readout rows [act_start : q_len]
    q_pos = torch.arange(k_len - q_len + act_start, k_len, device=logits_fp32.device, dtype=torch.
    float32).unsqueeze(1)
    k_pos = torch.arange(0, k_len, device=logits_fp32.device, dtype=torch.float32).unsqueeze(0)
    dist = (q_pos - k_pos).clamp_(min=0.0)
    if k_len > 32:
        dist[:, 0] = 0.0  # Pin <bos> token-0 attention sink

    # 3. 32-Token Local Syntax Guardband + Iso-Potential Cubic RG Plateau
    dist_long = (dist - 32.0).clamp_(min=0.0)
    c_l = layer_horizons.get(layer_idx, 0.54)
    horizon_tokens = c_l * max(float(k_len), calib_horizon)
    potential = (dist_long / max(horizon_tokens, 1.0)).clamp_(min=0.0, max=1.0).pow_(3.0)

    # 4. Dual-State Symmetric Boltzmann Extrapolation on isolated Global Logic Quorum
    for h in range(num_heads):
        head_slice = logits_fp32[:, h, act_start:, :]
        if (layer_idx, h) in target_heads:
            kappa = target_heads[(layer_idx, h)]
            delta_l = (beta * kappa * potential).unsqueeze(0)
            s_free = torch.softmax(head_slice + delta_l, dim=-1)
            s_heavy = torch.softmax(head_slice - delta_l, dim=-1)
            s_super = ((1.0 + eta) * s_free - eta * s_heavy).clamp_(min=0.0)
            logits_fp32[:, h, act_start:, :] = s_super / (s_super.sum(dim=-1, keepdim=True) + 1e-9)
        else:
            logits_fp32[:, h, act_start:, :] = torch.softmax(head_slice, dim=-1)

    return logits_fp32
```